Form S-6 Interactive Grid and Wind Fields

Sensitivity and Uncertainty Analysis over Southern Florida

Paul Fishwick and Claude Code

June 18, 2026

1 Purpose

This project implements the grid and wind-field machinery underlying *Form S-6 (Hypothetical Events for Sensitivity and Uncertainty Analysis)* of the Florida Commission on Hurricane Loss Projection Methodology Report of Activities (ROA), which supports *Standard S-3 (Uncertainty Analysis for Hurricane Model Output).* The deliverable is an interactive browser application (HTML/CSS/JavaScript + Leaflet) that draws the official 21×40 grid over southern Florida, runs selectable hurricane wind-field models over it, and performs the sensitivity (SRC) and uncertainty (EPR) analyses the ROA prescribes.

2 The grid and storm track

The grid (ROA Figs. 3–4) has $21 \times 40 = 840$ vertices at $\sim$3 statute-mile spacing: east–west $E = 0, 3, \ldots, 117$ (40 columns) and north–south $N = -15, -12, \ldots, 45$ (21 rows). Of the 840 vertices, **682 are land** and 158 water, taken directly from the `Land-Water ID` worksheet of `FormS6Input.xlsx`. The point $(0, 0)$ is the storm centre at $t = 0$, nine miles east of the landfall location (25.8611°N, 80.1196°W); the storm tracks due west, $(0, 0) \rightarrow (117, 0)$, over 12 hours. Figure 1 shows the rendered grid (land vs. water), the track, and the landfall marker.

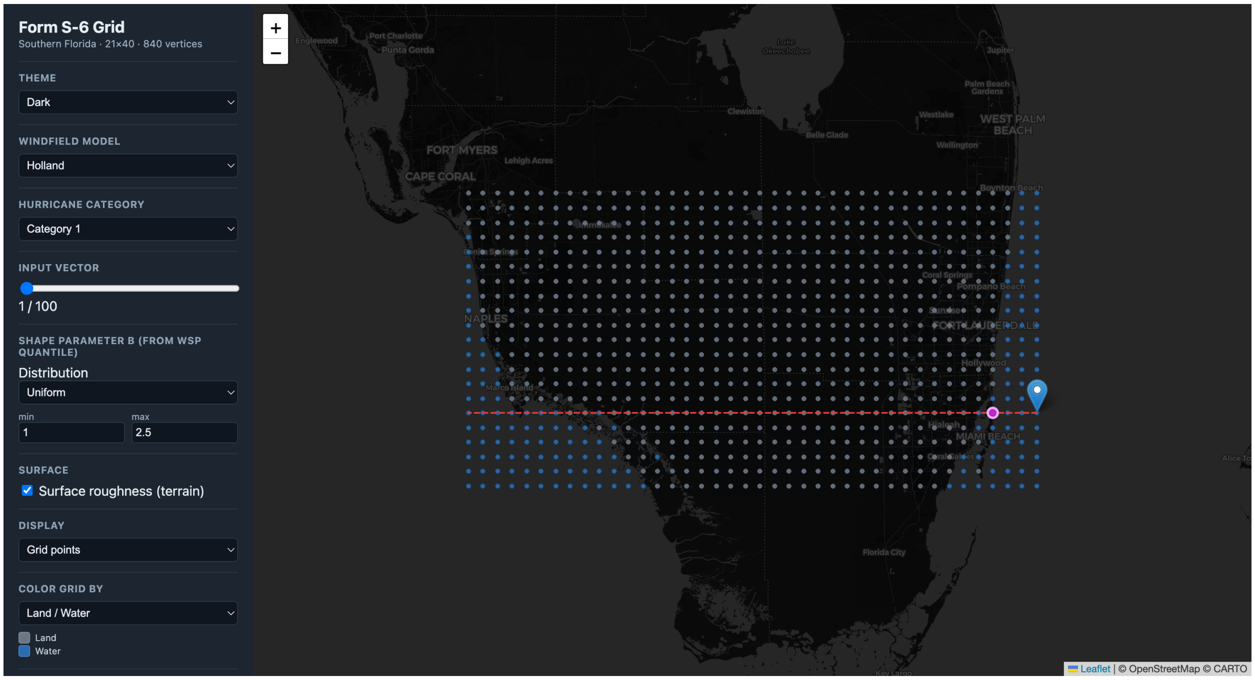

Figure 1: The 840-vertex Form S-6 grid over southern Florida: land (grey) and water (blue) vertices, the due-west storm track (dashed), the $t=0$ centre, and the landfall marker.

3 Wind-field models

Three wind-field models are provided, each evaluated at all 840 vertices. **Powell** is an iterative boundary-layer PDE (slab) solver; **Holland** is the closed-form gradient-wind profile; **Willoughby** is a piecewise power-law profile anchored to the Holland $V_{\max}$. All three include translation asymmetry from the storm motion. Powell is precomputed in Python (PyTorch/MPS, ~2.3 s per solve); Holland and Willoughby are closed-form and computed live in the browser.

Per-vector inputs come from the `SA all Variables` worksheet: central pressure CP, radius of maximum wind Rmax, translation speed VT, wind-field shape parameter WSP, conversion factor CF, and far-field pressure FFP, for 100 vectors in each of categories 1, 3, and 5. The pressure deficit is $\Delta p = \mathrm{FFP} - \mathrm{CP}$. WSP is supplied as a quantile $p \in [0,1]$ and mapped to the Holland shape parameter B through a user-selectable distribution (default Uniform$[1.0, 2.5]$, $B = B_{\min} + p\,(B_{\max} - B_{\min})$).

Conversion factor (CF). The ROA CF variable converts modelled gradient winds to surface winds using a three-zone radial rule (ROA pp. 184–185): for $r < R_{\max}$, $\mathrm{CF}_{\mathrm{eff}} = \mathrm{CF} \cdot r/R_{\max}$; for $R_{\max} < r < 3R_{\max}$, $\mathrm{CF}_{\mathrm{eff}} = \mathrm{CF} - 0.1\,(r - R_{\max})/(2R_{\max})$; and for $r > 3R_{\max}$, $\mathrm{CF}_{\mathrm{eff}} = \mathrm{CF} - 0.1$. The models are run at gradient level ($\beta_{10} = 1$) and CF performs the gradient→surface conversion exactly as specified.

Peak-wind metric and time sampling. For each vector we record the per-vertex **12-hour peak surface wind**. The storm position is sampled finely ($\Delta t = 0.1$ h); coarse hourly sampling aliases the fast westward motion and produces a spurious "comb" artifact in the peak field, which fine sampling removes. Figures 2 and 3 show the peak field as coloured points and as filled contours.

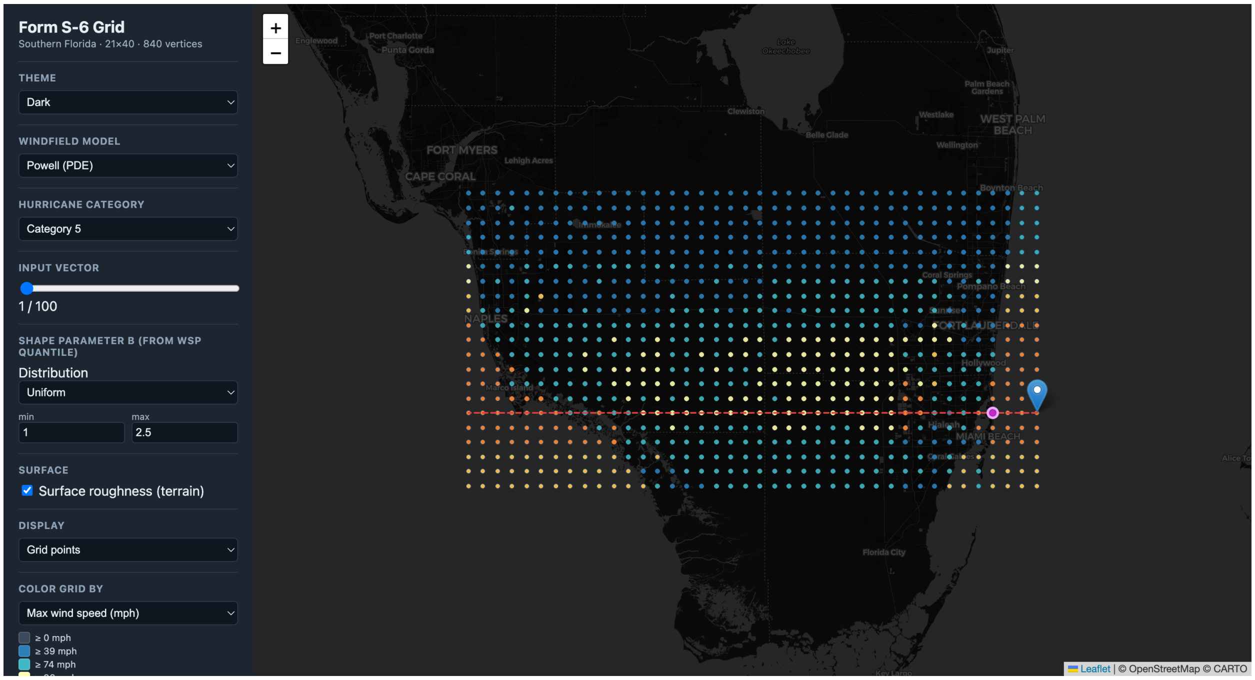

Figure 2: Powell peak surface wind (Category 5, vector 1) coloured per vertex, with surface roughness applied. Strongest winds lie north of the track—correct for a westward-moving Northern-Hemisphere cyclone, where translation adds to rotation on the north side.

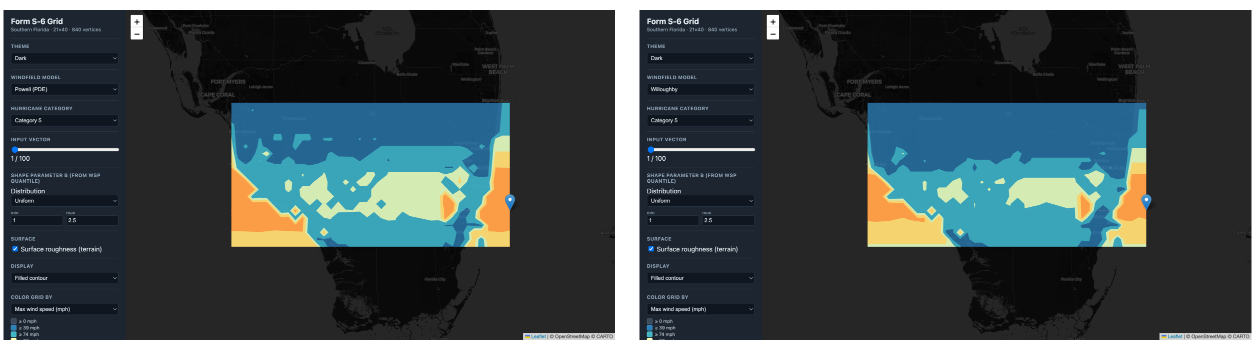

Figure 3: Filled-contour display (ROA Figs. 6–8 style) of the Category 5 peak field: Powell (left) and Willoughby (right). Contours are built on the 21×40 lattice with the wind-speed colour bands.

4 Surface roughness

Surface roughness (terrain) is applied as a per-vertex, wind-speed-independent multiplier on the final surface wind, toggleable in the interface. The roughness length z_0 at each vertex is sampled from **NLCD 2021 land cover** (fetched as a properly georeferenced GeoTIFF from the MRLC WCS) via a published land-cover$\rightarrow z_0$ table, and converted from marine to terrain exposure with the gradient-tied **log-law** (Vickery et al. 2009 / ESDU):

$$\text{factor} = \frac{\ln(z_{\text{ref}}/z_{0,\text{land}})/\ln(z_g/z_{0,\text{land}})}{\ln(z_{\text{ref}}/z_{0,\text{marine}})/\ln(z_g/z_{0,\text{marine}})},$$

with $z_{\text{ref}} = 10\,\text{m}$ and gradient height $z_g = 500\,\text{m}$; water returns ≈ 1. The land-mean factor is ≈ 0.67 (urban ~ 0.50, wetland ~ 0.70). This replaced an earlier heuristic that relied on an un-georeferenced JPEG.

5 Sensitivity and uncertainty analysis

Following Iman, Johnson, and Schroeder (2001), as the ROA cites:

- **Sensitivity (SRC).** Standardized regression coefficients from regressing the standardized output on the six standardized inputs over the 100 SA vectors, per category. Sign gives direction, magnitude gives influence (ROA Fig. 9).
- **Uncertainty (EPR).** Expected percentage reduction in output variance per input, $\text{EPR}_i = \text{SRC}_i^2 \times 100\%$ (variance share; valid for near-independent inputs, as the ROA design intends) (ROA Fig. 10).

The output metric is the **mean peak wind over the 682 land vertices** (a wind proxy; the loss-cost output of the full Form S-6 is future work). The computed SRCs reproduce the ROA's qualitative findings: WSP dominates for Category 1 while Rmax grows to dominate for Category 5, with CP negative and CF, FFP, Rmax positive (Figure 4).

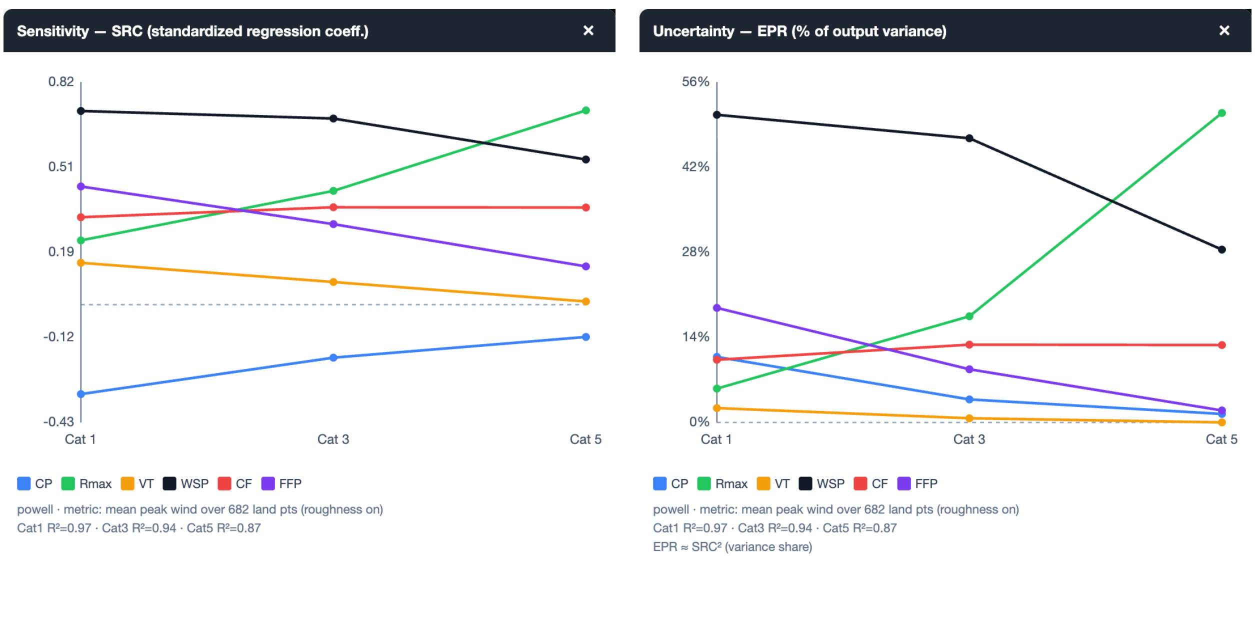

Figure 4: Sensitivity (SRC, left) and uncertainty (EPR, right) for the Powell model, plotted by hurricane category—the interactive analogues of ROA Figs. 9–10. The windows are independently movable and resizable.

6 Interface and reproducibility

The viewer offers model / category / input-vector selection, the WSP$\rightarrow B$ distribution control, a surface-roughness toggle, a points/filled-contour display toggle, light and dark themes (Figure 5), nearest-point hover (peak wind, coordinates, land/water, place name, and the active input parameters), and the floating Sensitivity/Uncertainty windows.

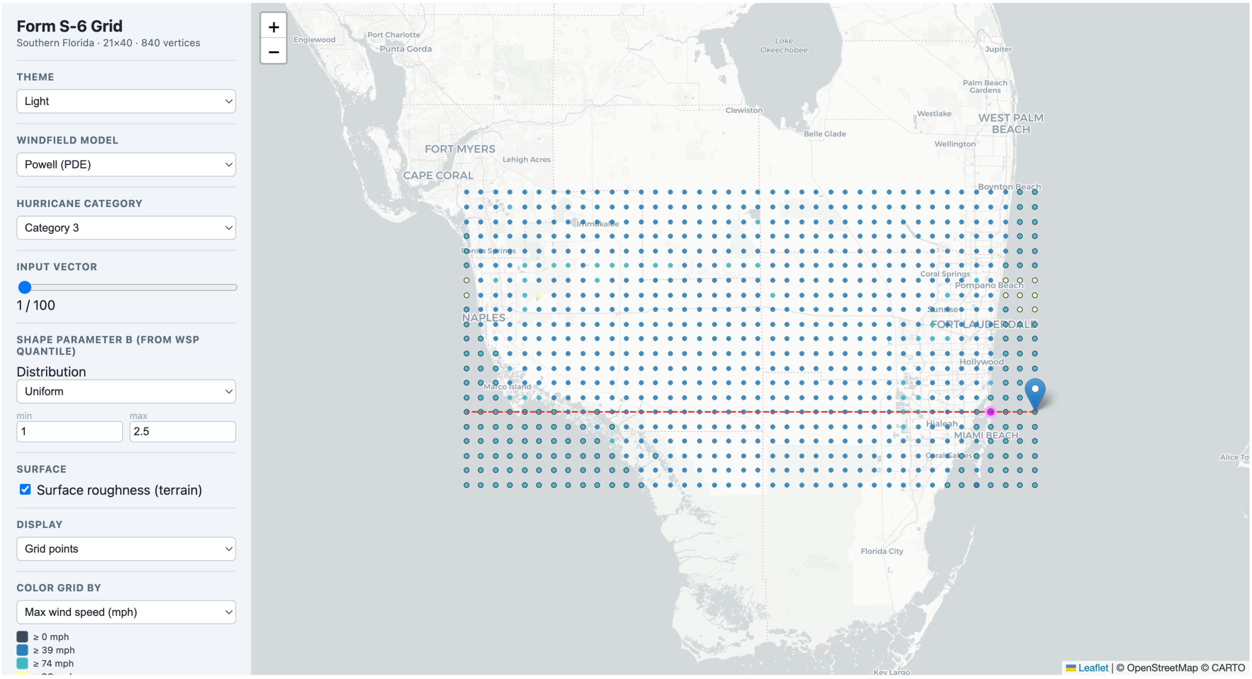

Figure 5: Light theme, Powell Category 3 peak-wind points.

All data artifacts are reproducible via `pipeline/build_all.sh` (`grid.json`, place names, NLCD fetch, `roughness.json`, `inputs.json`, `powell.json`). The app is served with `./start` at `http://localhost:8012/web/index.html`. Figures in this report are captured by `docs/capture_figures.py` (Selenium driving headless Chrome against the running viewer).

References

[1] Florida Commission on Hurricane Loss Projection Methodology (2025). *Report of Activities.* Standard S-3 and Form S-6.

[2] Iman, R. L., Johnson, M. E., and Schroeder, T. A. (2001). *Professional Team Demonstration Uncertainty/Sensitivity Analysis.* `https://fchlpm.sbafla.com/media/tepang2l/ua-sa-demo.pdf`

[3] Holland, G. J. (1980). An analytic model of the wind and pressure profiles in hurricanes. *Monthly Weather Review*, 108, 1212–1218.

[4] Willoughby, H. E., Darling, R. W. R., and Rahn, M. E. (2006). Parametric representation of the primary hurricane vortex. Part II: A new family of sectionally continuous profiles. *Monthly Weather Review*, 134, 1102–1120.

[5] Powell, M. D., Houston, S. H., and Reinhold, T. A. (1996). Hurricane Andrew's landfall in South Florida. Part I: Standardizing the surface wind observations. *Weather and Forecasting*, 11, 304–328.

[6] Vickery, P. J., Wadhera, D., Powell, M. D., and Chen, Y. (2009). A hurricane boundary layer and wind field model for use in engineering applications. *Journal of Applied Meteorology and Climatology*, 48, 381–405.

[7] Kaplan, J., and DeMaria, M. (1995). A simple empirical model for predicting the decay of tropical cyclone winds after landfall. *Journal of Applied Meteorology*, 34, 2499–2512.

[8] ESDU (1985). *Characteristics of atmospheric turbulence near the ground. Part II: Single point data for strong winds (neutral atmosphere)*. ESDU 85020.

[9] Multi-Resolution Land Characteristics Consortium (2023). *NLCD 2021 Land Cover (CONUS)*. `https://www.mrlc.gov`.

[10] U.S. Environmental Protection Agency. *AERSURFACE User's Guide*, land-cover to surface-roughness (z_0) tables.